

KRISH RAY *Software Engineer*

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Profile

Aspiring Software Engineer with a strong academic record and dual international internship experience, specialising in Python, Full-Stack development, Robotics, and AI/ML systems. Proven expertise in architecting scalable applications, developing autonomous robotics solutions, and deploying production-grade systems. Adept at uniting advanced software engineering with AI-driven automation to deliver high-impact, real-world solutions.

Professional Experience

06/2025 – 08/2025
Zittau, Germany

Hochschule Zittau/Goerlitz (IPM)

Research Intern – Robotics & AI

- Engineered a modular, production-grade autonomous robotics system (LIMO Pro + MyCobot280 + YOLOv11, ROS2) orchestrating a distributed network of custom nodes for advanced SLAM mapping, real-time 3D perception, and precision robotic manipulation, unified through a state-machine controller to enable fully automated exploration, high-fidelity object detection, intelligent pick-and-place, and seamless return-to-base workflows via a single-click deployment pipeline.

06/2024 – 08/2024
Indore, India

TechSolve LLP

Software Developer Intern

- Developed and deployed AI-powered and web-based solutions, including a Django-based resume builder and advanced web-scraping automation pipelines using Selenium and BeautifulSoup4, enhancing data processing efficiency by 70%; engineered and optimised backend APIs, integrated relational databases, and delivered scalable, production-ready applications.

Skills

Programming & CS: Python, Java, C/C++, JavaScript, HTML/CSS, SQL, MATLAB, GDScript
OOPS, DSA, OS, CSA, Automata, Cryptography

Web & APIs: FastAPI, Django REST Framework, Pydantic • React, Next.js, Astro, RESTful APIs

AI/ML & Agents: CrewAI, LangGraph, LangChain • YOLOv8, OpenCV, Transformers, NumPy, Pandas, Matplotlib

Robotics & Simulation: ROS2 (Foxy), TF2, Cartographer, Nav2, AMCL, Gazebo, RViz • MyCobot 280 control, SLAM, Navigation, autonomous exploration

Databases, Cloud & Tools: MySQL, PostgreSQL, SQLite • AWS • Git/GitHub, Linux, AutoCAD, STL workflows • Agile, Leadership, Cross-cultural collaboration, Technical Documentation (Docusaurus)

Projects

06/2025 – 07/2025

DogBot

Built a **ROS2 Foxy-based autonomous mobile manipulator** integrating LIMO Pro, MyCobot 280 arm, Orbbec DaBai camera, and YOLOv8 object detection, achieving **>95% pick-and-place accuracy** with SLAM mapping, waypoint navigation, depth-to-map transformation, and interrupt-safe mission control.

05/2025 – 06/2025

Agentic AI Orchestrator

Designed a multi-agent AI workflow system integrating **LangGraph**, **CrewAI**, Autogen and automation pipelines for autonomous task planning, execution, and monitoring, enabling scalable AI-driven decision-making.

06/2024 – 07/2024

AI Resume Builder & Full-Stack Web App

Developed a **Django-based application** leveraging **Meta LLaMA 3** for AI-generated resumes, featuring secure authentication, PDF export, responsive UI, and integrated PostgreSQL backend for data management and **automated web scraping pipelines** for skill and job-market keyword optimization.

06/2024 – 07/2024

Endless Space Combat Game

Designed and programmed a 2D space combat game in Godot using GDScript. Created custom physics, AI-driven enemy behavior, and dynamic difficulty progression.

References

Mradul Mishra, CTO, TechSolve LLP
hr@techsolve.com, +91 9669354334

Satpal Singh Kushwaha, Assistant Professor, Manipal University Jaipur
satpal.singh@jaipur.manipal.edu, +91 6376107467

Surbhi Sharma, Assistant Professor, Manipal University Jaipur
surbhi.sharma@jaipur.manipal.edu, +91 9462527657

Education

08/2023 – Present Jaipur, India	Manipal University Jaipur Bachelor of Technology in Computer Science and Engineering 8.72 CGPA
02/2021 – 02/2023 Indore, India	Prestige Public School 12TH GRADE 88.8%
02/2007 – 02/2021 Indore, India	St. Paul H.S. School 10TH GRADE 95.4%

Organisations

IAESTE LC MUJ Head of IT and Data Analytics	IEEE Robotics and Automation Society Head of Logistics	Manipal University Jaipur Class Representative(CR)
Rotaract Club Service Domain	NEXUS Senior Working Team	Association for Computing Machinery (ACM) Member

Languages

English C2	Hindi C2	German A1
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Certificates

Programming for Everybody (Python)  University of Michigan	C for Everyone: Programming Fundamentals  UC Santa Cruz	Object Oriented Programming (JAVA)  GeeksForGeeks
Database Programming with SQL Oracle	Database Foundations and Design Oracle	ROS2 Robotics Developer  Raymond Andrade

Awards

Dean's List
Manipal University Jaipur
Excellence in Academics(Highest Grade point average)